

Quantitative Discovery of Qualitative Information: A General Purpose Document Clustering Methodology

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joint work with
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The Problem: Discovery from Unstructured Text

- Examples: scholarly literature, news stories, medical information, blog posts, comments, product reviews, emails, social media updates, audio-to-text summaries, speeches, press releases, legal decisions, etc.
- 10 minutes of worldwide email = 1 LOC equivalent
- An essential part of discovery is **classification**: “one of the most central and generic of all our conceptual exercises. . . . the foundation not only for conceptualization, language, and speech, but also for mathematics, statistics, and data analysis. . . . Without classification, there could be no advanced conceptualization, reasoning, language, data analysis or, for that matter, social science research.” (Bailey, 1994).
- We focus on **cluster analysis**: discovery through (1) classification and (2) simultaneously inventing a classification scheme
- (We analyze text; our methods apply more generally)

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)
- $\text{Bell}(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $\text{Bell}(5) = 52$
- $\text{Bell}(100) \approx 10^{28} \times \text{Number of elementary particles in the universe}$
- Now imagine choosing the *optimal* classification scheme by hand
- Qualitative-only approaches are hopeless
- That we think of all this as astonishing ... is astonishing

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps, . . .
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **who knows?!**
 - The literature: **little guidance on when methods apply**
 - **Deriving such guidance: difficult or impossible**
 - (Perhaps true by definition in unsupervised learning: If we knew the DGP, we wouldn't be at the discovery stage.)

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model
- We do it **ex post** (by qualitative choice)
 - For discovery (our goal): No problem
 - For estimation & confirmation: more difficult or biased
- Complicated concepts are easier to define ex post:
 - “I know it when I see it” (Justice Stewart's definition of obscenity)
 - Anchoring Vignettes (on defining concepts by example)
- **But how to choose from an enormous list of clusterings?**

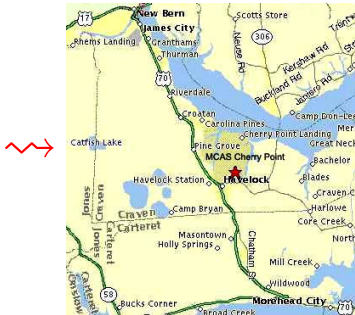
Our Idea: Meaning Through Geography

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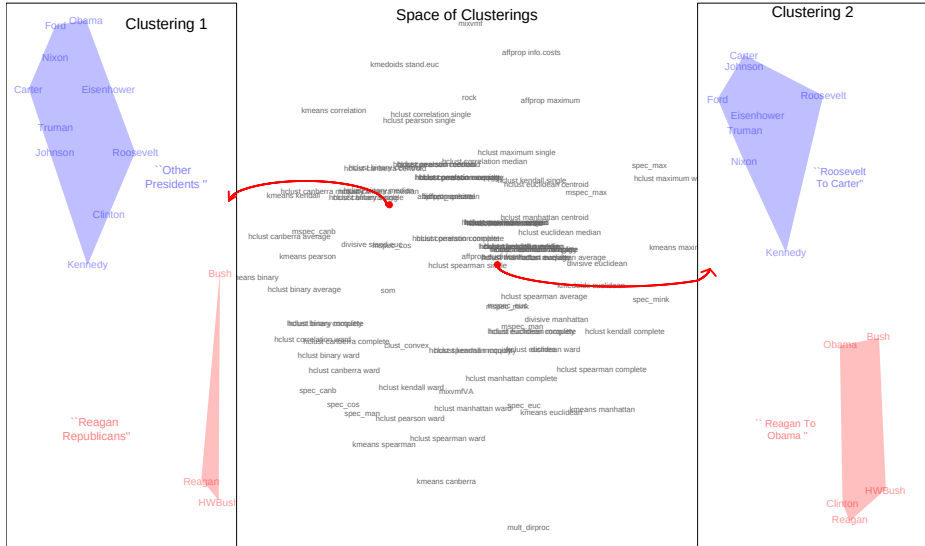
→ We develop a (conceptual) geography of clusterings

A New Strategy

- 1 Code text as numbers (in one or more of several ways)
- 2 Apply all existing clustering methods (that have been used by at least one person other than the author) to the data — each representing different substantive assumptions (<15 mins)
- 3 Develop an application-independent distance metric between clusterings
- 4 Create a metric space of clusterings, and a 2D projection
- 5 Introduce the local cluster ensemble to summarize any point, including points with no existing clustering
- 6 Propose a new animated visualization: use the local cluster ensemble to explore the space of clusterings (smoothly morphing from one into others)

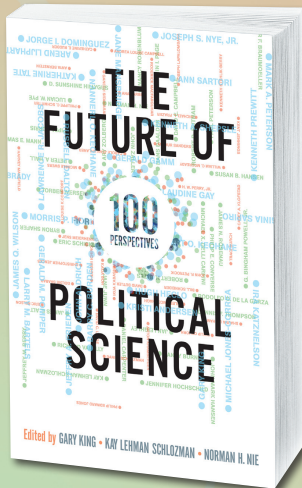
⇒ meaning revealed through a geography of clusterings

You choose one (or more), based on insight, discovery, useful information,...



Application-Independent Distance Metric: Axioms

- ① Clusterings with more **pairwise document agreements** are closer (we prove: pairwise agreements encompass triples, quadruples, etc.)
 - ② **Invariance**: Distance is invariant to the number of documents (for any fixed number of clusters)
 - ③ **Scale**: the maximum distance is set to $\log(\text{num clusters})$
- ⇒ **Only one measure satisfies all three** (the “variation of information”)



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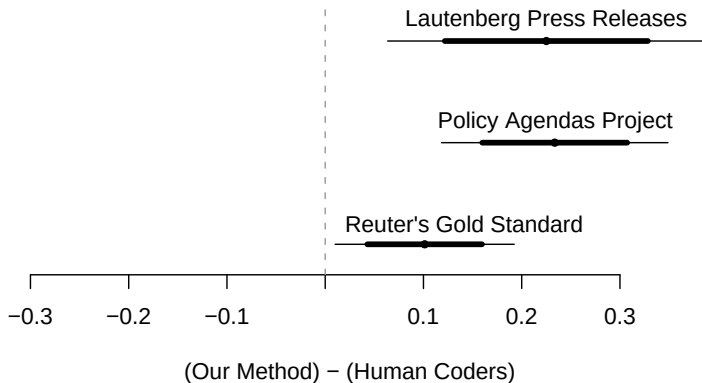
- Scale: (1) unrelated, (2) loosely related, or (3) closely related
- Table reports: mean(scale)

Pairs from	Overall Mean	Evaluator 1	Evaluator 2
Random Selection	1.38	1.16	1.60
Hand-Coded Clusters	1.58	1.48	1.68
Hand-Coding	2.06	1.88	2.24
Machine	2.24	2.08	2.40

p.s. The hand-coders did the evaluation!

Cluster Quality Experiments

Scale: $\text{mean}(\text{within clusters}) - \text{mean}(\text{between clusters})$



Lautenberg: 200 Senate Press Releases (appropriations, economy, education, tax, veterans, ...)

Policy Agendas: 213 quasi-sentences from Bush's State of the Union (agriculture, banking & commerce, civil rights/liberties, defense, ...)

Reuter's: financial news (trade, earnings, copper, gold, coffee, ...): "gold

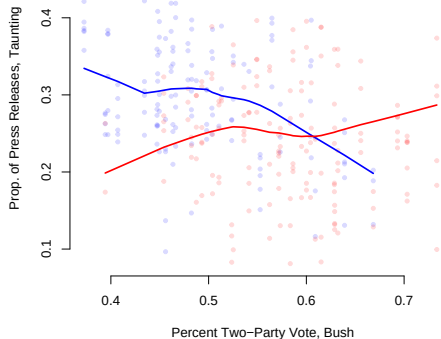
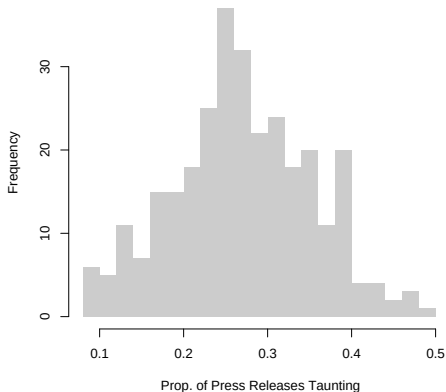
What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - ① Advertising
 - ② Credit Claiming
 - ③ Position Taking
- We find one more: **Partisan Taunting**
 - "Senator Lautenberg Blasts Republicans as 'Chicken Hawks' " [Government Oversight]
 - "The scopes trial took place in 1925. Sadly, President Bush's veto today shows that we haven't progressed much since then." [Healthcare]
 - "John Kerry had enough conviction to sign up for the military during wartime, unlike the Vice President, who had a deep conviction to avoid military service" [Government Oversight]
 - ~> **Is this what "party in government" means?**

Partisan Taunting Hypothesis Verification

Discovered in $n = 200$ (1 senator); confirmed in $n = 64,033$ (301 senator-years)



Prevalence of partisan taunting (it happens a lot!)

Partisan taunting occurs more in lopsided districts

↪ Conflict: deliberation (no taunting) v. reflection (lopsided districts)

More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- For each: created 2 clusterings from each of 3 methods, including ours
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons
- Both cases a Condorcet winner:

“Immigration”:

Our Method 1 → vMF 1 → vMF 2 → Our Method 2 → K-Means 1 → K-Means 2

“Genetic testing”:

Our Method 1 → {Our Method 2, K-Means 1, K-means 2} → Dir Proc. 1 → Dir Proc. 2

- **Intended contributions:**

- An encompassing cluster analytic approach for discovery
- A new approach to evaluating results in unsupervised learning
- Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Future research:**

- Advancing our approach: (1) $>2D$ exploration, (2) alternative visualizations of the space of clusterings, (3) including more methods
- Evaluating new individual methods: (1) distance from existing methods and their averages, (2) usefulness of discoveries in given data sets.

For more information:

<http://GKing.Harvard.edu>